

Assignment 7: GLMs (Linear Regressios, ANOVA, & t-tests)

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on generalized linear models.

Directions

1. Rename this file `<FirstLast>_A07_GLMs.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change “Student Name” on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure to **answer the questions** in this assignment document.
5. When you have completed the assignment, **Knit** the text and code into a single PDF file.

!! AI Statement: I did not use AI to complete this assignment. However, I use Grammarly to correct my grammar errors in writing the response for my paragraph answers, summary for my analysis AI generated code should NOT be used in this assignment

Set up your session

1. Set up your session. Check your working directory. Load the tidyverse, agricolae and other needed packages. Import the *raw* NTL-LTER raw data file for chemistry/physics (NTL-LTER_Lake_ChemistryPhysics_Raw.csv). Set date columns to date objects.
2. Build a ggplot theme and set it as your default theme.

```
#1 Load library packages and and import data for analysis
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.1      v stringr    1.5.2
## v ggplot2    4.0.0      v tibble     3.3.0
## v lubridate  1.9.4      v tidyr      1.3.1
## v purrr      1.1.0
```

```
## -- Conflicts ----- tidyverse_conflicts() --
```

```
## x dplyr::filter() masks stats::filter()
```

```
## x dplyr::lag()     masks stats::lag()
```

```
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(agricolae)
library(lubridate)
library(here)
```

```
## here() starts at /home/guest/EDE_FALL 2025
```

```
library(ggplot2)
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```
# get file path
getwd()
```

```
## [1] "/home/guest/EDE_FALL 2025"
```

```
# Load data data
```

```
Raw_NTLLTER <- read.csv (
  file = here("Data/Raw/NTL-LTER_Lake_ChemistryPhysics_Raw.csv"),
  stringsAsFactors = TRUE)
```

```
# Reading date as date
```

```
Raw_NTLLTER$sampledte <- mdy(Raw_NTLLTER$sampledte)
```

```
#2 Themes for my plots and graphs
```

```
Lab07_them <- theme_bw(base_size = 12) +
  theme(
    plot.title = element_text(hjust = 0.5, face = "bold", size = 14),
    axis.title = element_text(size = 11, face = "italic"),
    panel.grid.minor = element_blank(),
    legend.position = "bottom"
  )
theme_set(Lab07_them)
```

Simple regression

Our first research question is: Does mean lake temperature recorded during July change with depth across all lakes?

3. State the null and alternative hypotheses for this question: > Answer: H0: There might be no relationship between temperature and depth in July for all the lakes Ha: there might be a relationship between temperature and depth in July for all the lakes

```

# Find the date for July by using filtering tool. July count as number 7 in this case

data_july <- Raw_NTLTER %>%
  filter(month(sampledate) == 7 )
unique(month(data_july$sampledate))
nrow(data_july)

# Clean the selected data in month 7 with null value
clean_data7 <- data_july %>%
  filter(!is.na(depth), !is.na(temperature_C))
nrow(data_july) - nrow(clean_data7)

# Scatter plot the data into graph

ggplot(clean_data7, aes(x = depth, y = temperature_C)) +
  geom_point(alpha = 0.3) +
  geom_smooth(method = "lm") +
  labs(title = "Depth Level in July vs Temperature"
        , x = "Depth (m)",
        y = "Temperature (C)")

```

4. Wrangle your NTL-LTER dataset with a pipe function so that the records meet the following criteria:

- Only dates in July.
- Only the columns: `lakename`, `year4`, `daynum`, `depth`, `temperature_C`
- Only complete cases (i.e., remove NAs)

5. Visualize the relationship among the two continuous variables with a scatter plot of temperature by depth. Add a smoothed line showing the linear model, and limit temperature values from 0 to 35 °C. Make this plot look pretty and easy to read.

```

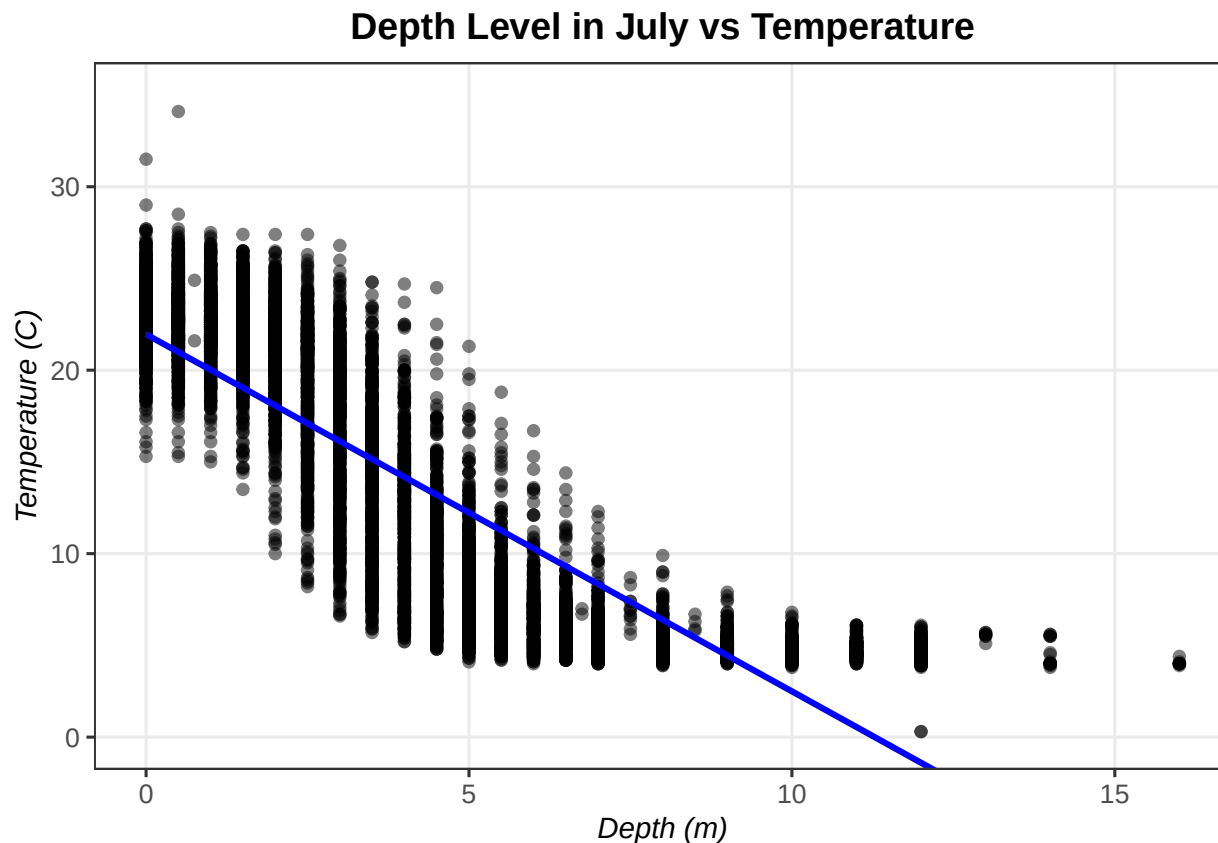
#4 July only dates
Dates_July <- Raw_NTLTER %>%
  filter(month(sampledate) == 7 ) %>%
select(lakename, year4, daynum, depth, temperature_C) %>%
  filter(complete.cases())

#5 Relationship plot

ggplot(Dates_July, aes(x = depth, y = temperature_C)) +
  geom_point(alpha = 0.5, color = "black") +
  geom_smooth(method = "lm", color = "blue") +
  coord_cartesian(ylim = c(0, 35)) +
  labs(title = "Depth Level in July vs Temperature",
        x = "Depth (m)",
        y = "Temperature (C)")

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



6. Interpret the figure. What does it suggest with regards to the response of temperature to depth? Do the distribution of points suggest about anything about the linearity of this trend?

Answer: The graph suggest that temperature is decreasing when going down deeper into the lake. Also the plotted points is curve meaning the suggestiing relationship may not be absolute linear.

7. Perform a linear regression to test the relationship and display the results.

```
#7
temp_depth_model <- lm(temperature_C ~ depth, data = Dates_July)
summary(temp_depth_model)
```

```
##
## Call:
## lm(formula = temperature_C ~ depth, data = Dates_July)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.5173  -3.0192   0.0633   2.9365  13.5834
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  21.95597    0.06792   323.3  <2e-16 ***
```

```
## depth          -1.94621    0.01174  -165.8   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.835 on 9726 degrees of freedom
## Multiple R-squared:  0.7387, Adjusted R-squared:  0.7387
## F-statistic: 2.75e+04 on 1 and 9726 DF,  p-value: < 2.2e-16
```

8. Interpret your model results in words. Include how much of the variability in temperature is explained by changes in depth, the degrees of freedom on which this finding is based, and the statistical significance of the result. Also mention how much temperature is predicted to change for every 1m change in depth.

Answer: During July, lake water becomes colder as depth increases. This relationship is very strong, with depth explaining about 73.8% of temperature changes. For every one meter of depth, temperature decreases by around 2°C

Multiple regression

Let's tackle a similar question from a different approach. Here, we want to explore what might the best set of predictors for lake temperature in July across the monitoring period at the North Temperate Lakes LTER.

9. Run an AIC to determine what set of explanatory variables (year4, daynum, depth) is best suited to predict temperature.
10. Run a multiple regression on the recommended set of variables.

#9

```
full_model <- lm(temperature_C ~ year4 + daynum + depth, data = Dates_July)
step_result <- step(full_model)
```

```
## Start:  AIC=26065.53
## temperature_C ~ year4 + daynum + depth
##
##           Df Sum of Sq    RSS   AIC
## <none>                 141687 26066
## - year4      1         101 141788 26070
## - daynum     1        1237 142924 26148
## - depth      1       404475 546161 39189
```

```
best_model <- step_result
summary(best_model)
```

```
##
## Call:
## lm(formula = temperature_C ~ year4 + daynum + depth, data = Dates_July)
##
```

```
## Residuals:
##      Min       1Q   Median       3Q      Max
## -9.6536 -3.0000  0.0902  2.9658 13.6123
##
## Coefficients:
##              Estimate Std. Error  t value Pr(>|t|)
## (Intercept) -8.575564   8.630715  -0.994  0.32044
## year4        0.011345   0.004299   2.639  0.00833 **
## daynum       0.039780   0.004317   9.215 < 2e-16 ***
## depth       -1.946437   0.011683 -166.611 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.817 on 9724 degrees of freedom
## Multiple R-squared:  0.7412, Adjusted R-squared:  0.7411
## F-statistic: 9283 on 3 and 9724 DF,  p-value: < 2.2e-16
```

#10

11. What is the final set of explanatory variables that the AIC method suggests we use to predict temperature in our multiple regression? How much of the observed variance does this model explain? Is this an improvement over the model using only depth as the explanatory variable?

Answer: The AIC method suggests using all three explanatory variables: year4, daynum, and depth to predict temperature. This model explains 74.12% of the observed variance in temperature 0.7412. This is an improvement over the model using only depth, which explained 73.87% of the variation an increase of about 0.25%.

Analysis of Variance

12. Now we want to see whether the different lakes have, on average, different temperatures in the month of July. Run an ANOVA test to complete this analysis. (No need to test assumptions of normality or similar variances.) Create two sets of models: one expressed as an ANOVA models and another expressed as a linear model (as done in our lessons).

#12

```
anova_model <- aov(temperature_C ~ lakenname, data = Dates_July)
lm_model <- lm(temperature_C ~ lakenname, data = Dates_July)
summary(anova_model)

##              Df Sum Sq Mean Sq F value Pr(>F)
## lakenname      8  21642   2705.2     50 <2e-16 ***
## Residuals    9719 525813     54.1
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

summary(lm_model)
```

```
##
## Call:
## lm(formula = temperature_C ~ lakename, data = Dates_July)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.769  -6.614  -2.679   7.684  23.832
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)      17.6664     0.6501  27.174 < 2e-16 ***
## lakenameCrampton Lake      -2.3145     0.7699  -3.006 0.002653 **
## lakenameEast Long Lake     -7.3987     0.6918 -10.695 < 2e-16 ***
## lakenameHummingbird Lake   -6.8931     0.9429  -7.311 2.87e-13 ***
## lakenamePaul Lake         -3.8522     0.6656  -5.788 7.36e-09 ***
## lakenamePeter Lake        -4.3501     0.6645  -6.547 6.17e-11 ***
## lakenameTuesday Lake     -6.5972     0.6769  -9.746 < 2e-16 ***
## lakenameWard Lake         -3.2078     0.9429  -3.402 0.000672 ***
## lakenameWest Long Lake    -6.0878     0.6895  -8.829 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 7.355 on 9719 degrees of freedom
## Multiple R-squared:  0.03953,    Adjusted R-squared:  0.03874
## F-statistic:    50 on 8 and 9719 DF,  p-value: < 2.2e-16
```

13. Is there a significant difference in mean temperature among the lakes? Report your findings.

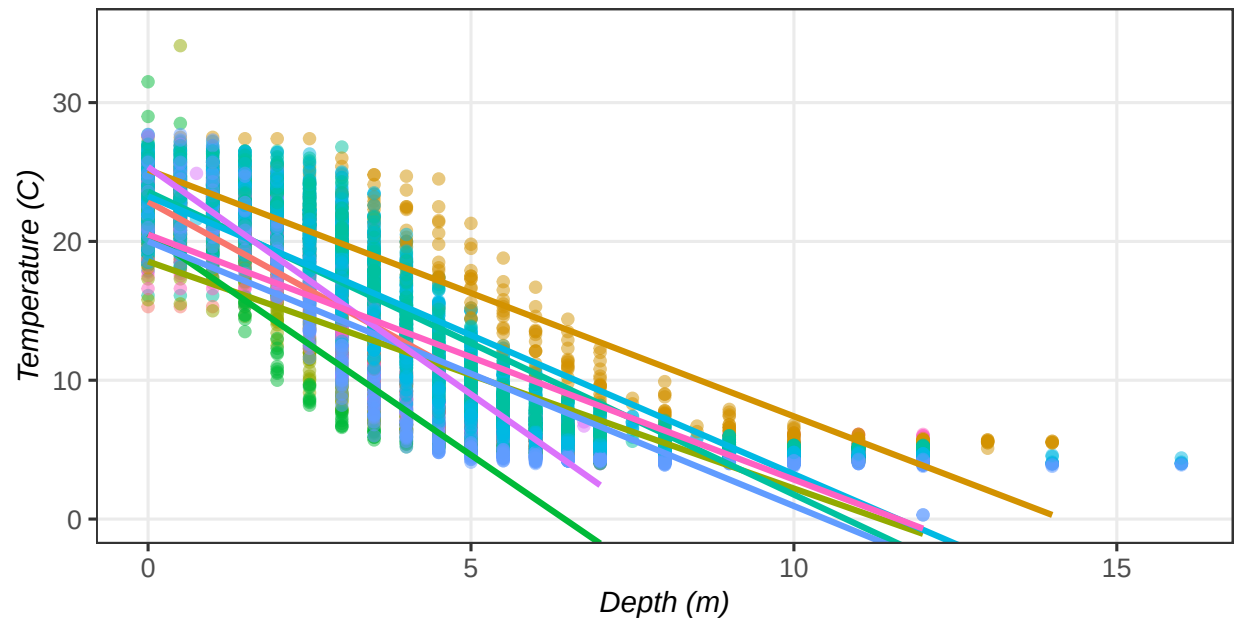
Answer: Yes, there are significant differences in average July temperatures between the lakes < 0.001.

14. Create a graph that depicts temperature by depth, with a separate color for each lake. Add a `geom_smooth` (method = "lm", se = FALSE) for each lake. Make your points 50 % transparent. Adjust your y axis limits to go from 0 to 35 degrees. Clean up your graph to make it pretty.

```
#14.
ggplot(Dates_July, aes(x = depth, y = temperature_C, color = lakename)) +
  geom_point(alpha = 0.5) +
  geom_smooth(method = "lm", se = FALSE) +
  coord_cartesian(ylim = c(0, 35)) +
  labs(title = "Temperature vs Depth by Lake",
       x = "Depth (m)",
       y = "Temperature (C)")
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Temperature vs Depth by Lake



name
 Central Long Lake East Long Lake Paul Lake Tuesday Lake West Long Lake
 Crampton Lake Hummingbird Lake Peter Lake Ward Lake

15. Use the Tukey's HSD test to determine which lakes have different means.

#15

```
tukey_result <- TukeyHSD(anova_model)
tukey_result
```

```
## Tukey multiple comparisons of means
## 95% family-wise confidence level
##
## Fit: aov(formula = temperature_C ~ lakename, data = Dates_July)
##
## $lakename
##
```

	diff	lwr	upr	p adj
Crampton Lake-Central Long Lake	-2.3145195	-4.7031913	0.0741524	0.0661566
East Long Lake-Central Long Lake	-7.3987410	-9.5449411	-5.2525408	0.0000000
Hummingbird Lake-Central Long Lake	-6.8931304	-9.8184178	-3.9678430	0.0000000
Paul Lake-Central Long Lake	-3.8521506	-5.9170942	-1.7872070	0.0000003
Peter Lake-Central Long Lake	-4.3501458	-6.4115874	-2.2887042	0.0000000
Tuesday Lake-Central Long Lake	-6.5971805	-8.6971605	-4.4972005	0.0000000
Ward Lake-Central Long Lake	-3.2077856	-6.1330730	-0.2824982	0.0193405
West Long Lake-Central Long Lake	-6.0877513	-8.2268550	-3.9486475	0.0000000
East Long Lake-Crampton Lake	-5.0842215	-6.5591700	-3.6092730	0.0000000
Hummingbird Lake-Crampton Lake	-4.5786109	-7.0538088	-2.1034131	0.0000004
Paul Lake-Crampton Lake	-1.5376312	-2.8916215	-0.1836408	0.0127491
Peter Lake-Crampton Lake	-2.0356263	-3.3842699	-0.6869828	0.0000999

## Tuesday Lake-Crampton Lake	-4.2826611	-5.6895065	-2.8758157	0.0000000
## Ward Lake-Crampton Lake	-0.8932661	-3.3684639	1.5819317	0.9714459
## West Long Lake-Crampton Lake	-3.7732318	-5.2378351	-2.3086285	0.0000000
## Hummingbird Lake-East Long Lake	0.5056106	-1.7364925	2.7477137	0.9988050
## Paul Lake-East Long Lake	3.5465903	2.6900206	4.4031601	0.0000000
## Peter Lake-East Long Lake	3.0485952	2.2005025	3.8966879	0.0000000
## Tuesday Lake-East Long Lake	0.8015604	-0.1363286	1.7394495	0.1657485
## Ward Lake-East Long Lake	4.1909554	1.9488523	6.4330585	0.0000002
## West Long Lake-East Long Lake	1.3109897	0.2885003	2.3334791	0.0022805
## Paul Lake-Hummingbird Lake	3.0409798	0.8765299	5.2054296	0.0004495
## Peter Lake-Hummingbird Lake	2.5429846	0.3818755	4.7040937	0.0080666
## Tuesday Lake-Hummingbird Lake	0.2959499	-1.9019508	2.4938505	0.9999752
## Ward Lake-Hummingbird Lake	3.6853448	0.6889874	6.6817022	0.0043297
## West Long Lake-Hummingbird Lake	0.8053791	-1.4299320	3.0406903	0.9717297
## Peter Lake-Paul Lake	-0.4979952	-1.1120620	0.1160717	0.2241586
## Tuesday Lake-Paul Lake	-2.7450299	-3.4781416	-2.0119182	0.0000000
## Ward Lake-Paul Lake	0.6443651	-1.5200848	2.8088149	0.9916978
## West Long Lake-Paul Lake	-2.2356007	-3.0742314	-1.3969699	0.0000000
## Tuesday Lake-Peter Lake	-2.2470347	-2.9702236	-1.5238458	0.0000000
## Ward Lake-Peter Lake	1.1423602	-1.0187489	3.3034693	0.7827037
## West Long Lake-Peter Lake	-1.7376055	-2.5675759	-0.9076350	0.0000000
## Ward Lake-Tuesday Lake	3.3893950	1.1914943	5.5872956	0.0000609
## West Long Lake-Tuesday Lake	0.5094292	-0.4121051	1.4309636	0.7374387
## West Long Lake-Ward Lake	-2.8799657	-5.1152769	-0.6446546	0.0021080

16. From the findings above, which lakes have the same mean temperature, statistically speaking, as Peter Lake? Does any lake have a mean temperature that is statistically distinct from all the other lakes?

Answer: Peter Lake has similar temperatures to Paul Lake and Ward Lake. Given Paul Lake $p = 0.224$ and Ward Lake $p = 0.783$.

17. If we were just looking at Peter Lake and Paul Lake. What's another test we might explore to see whether they have distinct mean temperatures?

Answer: Based on the Tukey test that Peter Lake and Paul Lake are not significantly different $p = 0.224$, we could use a two-sample t-test to specifically compare just these two lakes and confirm their temperature similarity.

18. Wrangle the July data to include only records for Crampton Lake and Ward Lake. Run the two-sample T-test on these data to determine whether their July temperature are same or different. What does the test say? Are the mean temperatures for the lakes equal? Does that match your answer for part 16?

```
#18
crampton_ward <- Dates_July %>% filter(lakename %in% c("Crampton Lake", "Ward Lake"))
t_test_result <- t.test(temperature_C ~ lakename, data = crampton_ward)
t_test_result

##
## Welch Two Sample t-test
##
## data: temperature_C by lakename
```

```
## t = 1.1181, df = 200.37, p-value = 0.2649
## alternative hypothesis: true difference in means between group Crampton Lake and group Ward Lake is not equal to 0
## 95 percent confidence interval:
## -0.6821129  2.4686451
## sample estimates:
## mean in group Crampton Lake      mean in group Ward Lake
##              15.35189              14.45862
```

Answer: The t-test results show no significant difference between Crampton Lake 15.35°C and Ward Lake 14.46°C temperatures, with a p-value of 0.2649. This means should not include their mean temperatures different.